MIL-I-8276B(USAF)
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SUPERSEDING
MIL-I-8276A
26 July 1954

MILITARY SPECIFICATION

INDICATOR, RATE OF FLOW, FUEL, SYNCHRO STYLE, TYPE MA-2

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a single remote-indicating fuel flow rate indicator, in a 2-inch case, designated Type MA-2.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards, form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

Federal

PPP-B-601	Boxes, Wood, Cleated Plywood
PPP-B-636	Box, Shipping, Fiberboard

Military

MIL-P-116	Preservation, Method Of
MIL-I-7057	Indicators, Synchro, Aircraft, General Specification For
MIL-T-8275	Transmitter, Rate Of Flow, Synchro Style, Type MA-1
MIL-L-25142	Luminescent Material, Fluorescent

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Oklahoma City Air Logistics Center/MMEDO, Tinker AFB OK 73145-5990 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6620
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MILITARY

MIL-STD-129	Marking For Shipment And Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-2073-1	DOD Material Procedures For Development And Application Of Packing Requirements
MS33585	Pointers, Dial, Standard Design of Aircraft Instrument
MS33639	Cases, Instrument, Clamp-Mounted, Aircraft
MS33678	Connector, Receptacle, Electric Integral Mounting

(Copies of specifications, standards, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

American Society For Testing And Materials (ASTM)

ASTM D3951 Packaging, Commercial

(Application for copies should be addressed to: ASTM, 1916 Race St, Philadelphia PA 19103.)

(Nongovernment standards and other publications are normally available from the organization which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The requirements specified in Specification MIL-I-7057 are applicable as requirements of this specification. Additional requirements shall be as specified herein. In the event the requirements of the general specification and this specification conflict, the requirements of this specification shall govern.

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3.1 Indicator. The indicator shall operate on the synchro system specified in Specification MIL-I-7057.

3.2 Design and construction.

3.2.1 Calibration. Each indicator dial shall be graduated in accordance with figure 1. The calibration of fuel flow versus angular degrees shall be linear within $\pm 0^{o}$ 15' of arc, except at the changeover point.

3.2.1.1 Transmitter. Unless ohterwise specified, the indicator shall be designed to indicate, at a remote station, the rate of flow in pounds per hour of aviation fuel when used in conjunction with a fuel flow transmitter in accordance with Specification MIL-T-8275.

3.2.2 Pointer. The pointer shall conform to MS33585-8 except that the length of the pointer shall be such that the tip will overlap from 0.333 to 0.667 the length of the shortest graduation.

3.2.3 Fluorescent-luminescent markings. The following markings shall be finished in fluorescent-luminescent material per MIL-L-25142. The dimensions of the markings shall be as follows:

Markings	Height or length Inch ± 0.016	Width of line or graduation Inch ±0.005
Scale Numerals 0 through 6	0.156	0.031
Scale Numerals 8 through 24	0.094	0.020
Numbered 1,000-pound graduations 0 through 6	0.156	0.031
Numbered 1,000-pound graduations 6 through 24	0.156	0.020
Intermediate 1,000-pounds graduations	0.125	0.020
500-pound graduations	0.125	0.031
100-pound graduations	0.094	0.020
Lettering "PPH x 1000"	0.094	0.016
Lettering "FUEL FLOW"	0.125	0.020

3.2.3.1 Durable dull black. The marking "TYPE MA-2" shall be permanently and legibly marked on the dial in 0.047 inch letters. This marking and all other markings, not otherwise specified, shall be finished in durable dull black.

3.2.4 Case. The case shall be hermetically sealed and shall conform to Drawing MS33639. The dimensions "L" shall be 2.750 ±0.250.

3.2.5 Dial diameter. The dial shall have a minimum diameter of 1.750 inches as measured at the outside ends of the scale graduations.

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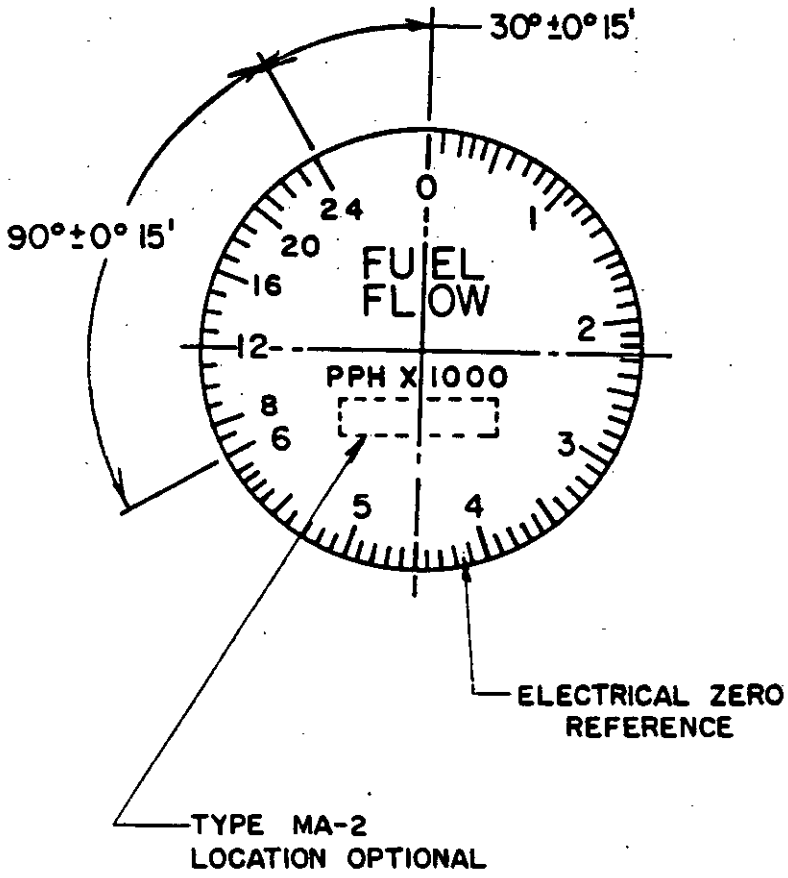


FIGURE 1. INDICATOR dial face.

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3.2.5.1 Electrical receptacle. The indicator electrical receptacle shall be in accordance with MS33678-145-2P.

3.3 Weight. The weight of the complete indicator shall not exceed 0.6 pound.

3.4 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with Standard MIL-STD-130.

3.5 Recycled and reclaimed materials. The use of recycled and reclaimed materials shall be encouraged to the maximum extent possible without jeopardizing the intended end use of the item.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as othwerwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3, 4 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsiblity of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.1.1 General. The sampling, inspection, and test procedures shall be in accordance with Specification MIL-I-7057.

4.2 Performance. The performance of the indicator shall be checked against tables I and II of this specification

4.3 Inspection of packaging. Test of methods of preservation and packaging shall be accomplished in acordance with Section 4 of MIL-P-116 to insure compliance with Section 5 of this specification.

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TABLE I. Indicator fuel flow rate tolerances.

	Tolerance PPH ±		
Flow rate PPH	Room temp	Low temp	High temp
400	20	30	20
1,000	20	30	20
2,000	20	30	20
3,000	20	30	20
4,000	20	30	20
5,000	20	30	20
6,000	20	30	20
8,000	200	300	200
12,000	200	300	200
16,000	200	300	200
20,000	200	300	200
24,000	200	300	200

TABLE II. Indicator test tolerances.

	Tolerance PPH ±	
Test	0-6,000 Scale	6,000-24,000 Scale
Frequency and Voltage variation	20	200
Position error at 5,000 PPH	20	—
Friction:		
Room temperature	40	300
High temperature	40	300
Low temperature	65	500
Vibration:		
Pointer oscillation	40	400
Pointer variation	40	400
Endurance:		
Response time 3 second max	40	300

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5. PACKAGING

5.1 Preservation. Preservation shall be level A, C, or Industrial, IAW MIL-STD-2073-1, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning. Indicator shall be cleaned in accordance with process D-4 of MIL-P-116.

5.1.1.2 Preservation application. Not applicable.

5.1.1.4 Unit packaging. Unless otherwise specified by the contracting activity, each indicator shall be packaged in quantity unit packs of one each in accordance with Method IC-1 of MIL-P-116. Each indicator shall be placed in a PPP-B-636 Fiberboard container weather resistant, with sufficient cushioning material between bag and unit container of a type, density, and thickness to insure shock transmission does not exceed peak values in G's established for the indicator when completed packs are subjected to the rough handling drop tests of MIL-P-116.

5.1.2 Level C. Each indicator shall be clean, dry, and individually packaged in a manner that will afford adequate protection against corrosion, deterioration, and physical damage during shipment from supply source to the first receiving activity.

5.1.3 Industrial. The Industrial preservation of indicator shall be in accordance with ASTM D3951.

5.2 Packing. Packing shall be level A, B, C or Industrial, IAW MIL-STD-2073-1, as specified (see 6.2).

5.2.1 Level A. Indicator packaged as specified in 5.1.1 shall be packed in shipping containers conforming to PPP-B-601, Styles A or B, Class overseas, unless otherwise specified by the contracting activity. Insofar as practical, exterior shipping container shall be of uniform shape, size, minimum tare and cube consistent with the protection required.

5.2.2 Level B. Indicator packaged as specified in 5.1.1 shall be packed in shipping containers conforming to PPP-B-636, class weather-resistant, unless otherwise specified by the contracting activity. Other requirements as specified in 5.2.1 apply.

5.2.3 Level C. Packing shall be applied which affords adequate protection during domestic shipment from the supply source to the first receiving activity for immediate use. This level shall conform to applicable carrier rules and regulations.

5.2.4 Industrial. The packaged indicator shall be packed in accordance with ASTM D3951.

5.3 Marking. In addition to any other markings required by the contract or order (see 6.2), interior and exterior containers shall be marked in accordance with MIL-STD-129.

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6. NOTES

6.1 Intended use. The Type MA-2 indicator covered by this specification is intended for use to indicate the rate of flow of fuel in pounds per hour consumed by an aircraft turbo-jet engine.

6.2 Ordering data. Procurement documents shall specify the following:

a. Title, number and data of this specification.

b. Selection of applicable levels of preservation, packaging and packing.

c. Unit quantities.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time set for opening of bids, qualified for inclusion in Qualified Products List (QPL No.) whether or not such products have actually been so listed by that date.

The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. The activity responsible for the Qualified Products List is Oklahoma City Air Logistics Center/MMEDO, Tinker AFB OK 73145-5990 and information pertaining to qualification of products may be obtained from that activity.

6.4 Subject term (key word) listing.

Fuel
Indicator
MIL-T-8275
Rate of Flow
Synchro
Type MA-2
0-24,000 PPH

6.5 Changes from previous issue. Asterisks (or vertical lines)are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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